Lesson notes · for you

Adversaries and noise

University, vision 4

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Rebuild from a code
- Fool a judge

Activity: In the lab

1. Noise in the middle
2. Noise at the end

Discussion

- Which failure is whose?

Check yourself

- Name the method that undoes noise.

Try it this week
Write the three methods and one failure each, from memory.

Worksheet

Adversaries and noise

University, vision 4

Name _____ Class _____ Date _____

1 **Which idea removes a little noise at a time?**

◯ Diffusion ◯ A single 3×3 edge filter ◯ A password hash

2 **A picture from a generator is evidence of what it shows.**

◯ True ◯ False

3 **Match the generator.**

Autoencoder	• • • •	Undo noise one step at a time
GAN	• • • •	One network tries to fool another
Diffusion	• • • •	Rebuild from a short code

4 **Order a diffusion story.**

☐ Repeat the step

☐ Learn a step that removes a little

☐ Add noise until the picture is gone

✎ **Why is one denoise step not a picture coming back?**